

Complex Engineering Problem

Course: Machine Learning

Topic: Support Vector Machines (SVM)

Problem:

You are given the following training dataset for a binary classification problem:

Sample	x_1	x_2	Class (y)
A	2	2	+1
B	4	4	+1
C	4	0	-1
D	0	4	-1

(a) Linear Separability & Visualization

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- Plot the data points on a 2D plane
- Determine whether the dataset is **linearly separable**
- Justify your answer

(b) Optimal Hyperplane

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Find the **maximum margin hyperplane** of the form:

$$w_1 x_1 + w_2 x_2 + b = 0$$

- Identify the **support vectors**
- Compute **w** and **b**
- Show all necessary steps

(c) Margin Calculation

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Compute the **margin** of the classifier:

$$\text{Margin} = \frac{2}{\|w\|}$$

(d) Non-Linear Extension

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Suppose the data becomes non-linearly separable.

- Explain how the **kernel trick** can be applied
- Write the decision function using a kernel

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x_i, x) + b$$

- Name two commonly used kernels and briefly explain them

(e) Design Considerations

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Discuss the following SVM design issues:

- Choice of kernel
- Regularization parameter (C)
- Overfitting vs underfitting trade-off